

Chapter 1, Problem 39CP

Problem

A 600-W TV receiver is turned on for 4 h with nobody watching it. If electricity costs 10 cents/kWh, how much money is wasted?

Step-by-step solution

Step 1 of 2

Consider the following data:

A 600-W TV receiver tuned for 4 h .

The cost of electricity is, 10 cents/kWh .

Calculate the power consumed for 4 h is as follows.

$$\begin{aligned}\text{power consumed for 4 hours} &= (600\text{-W})(4\text{ h}) \\ &= 2400\text{ Wh} \\ &= 2.4\text{ kWh}\end{aligned}$$

Step 2 of 2

Calculate the cost of electricity is as follows.

$$\begin{aligned}2.4\text{ kWh @ }10\text{ cents/kWh} &= (2.4\text{ kWh})(10\text{ cents/kWh}) \\ &= 24\text{ cents}\end{aligned}$$

Therefore, the money wasted is 24 cents .